Conceptual Q&A: CNNs and RNNs

CNN: Convolutional Neural Networks

Q: What is the purpose of the convolutional layer in a CNN?

A: The convolutional layer is responsible for automatically extracting spatial features from input data, like edges, textures, or patterns. It uses filters (kernels) that slide over the input to produce feature maps highlighting important visual cues.

Q: How does a convolutional layer work?

A: Each filter (a small matrix) performs element-wise multiplication with a portion of the input (called the receptive field), and the results are summed to form a single value in the output feature map. This operation is repeated over the whole input using a stride.

Q: What is the function of a pooling layer in CNNs?

A: Pooling layers reduce the spatial dimensions (width and height) of the feature maps. This helps in reducing the number of parameters, computation, and overfitting. It also makes the representation more robust to small translations.

Q: What are common types of pooling?

A: - Max pooling: Takes the maximum value in each patch.

- Average pooling: Takes the average of values in each patch.

Q: Why are CNNs preferred for image-related tasks?

A: CNNs efficiently capture spatial hierarchies and patterns due to local receptive fields, shared weights, and translation invariance, making them ideal for tasks like image classification, object detection, and segmentation.

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RNN: Recurrent Neural Networks

Q: What is a basic RNN and what problem does it solve?

A: A basic RNN is a type of neural network designed to handle sequential data by maintaining a memory of previous inputs through hidden states. It's commonly used in tasks like language modeling, time-series forecasting, and speech recognition.

Q: How does memory update work in a basic RNN?

A: At each time step t, the RNN updates its hidden state h_t using the current input x_t and the previous hidden state h_{t-1}:

h_t = tanh(W_h h_{t-1} + W_x x_t + b)

This way, the network retains some information about past inputs while processing new ones.

Q: What are the limitations of basic RNNs?

A: - Struggles with long-term dependencies due to vanishing or exploding gradients.

- Can forget older information when sequences are long.

Q: What are the advantages of RNNs?

A: - Designed to handle sequential and temporal data.

- Capable of modeling context and order in data like text and speech.

Q: How do modern RNN variants (like LSTM/GRU) address the limitations?

A: They include gating mechanisms to control what information is remembered or forgotten. For example,

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LSTMs use input, forget, and output gates to manage memory more effectively and handle longer sequences better.